M. TALMIZ UR REHMAN

Embedded Firmware & Hardware Design Engineer

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SUMMARY

Embedded firmware and hardware engineer with years across the full product lifecycle — firmware and BSP bring-up, PCB design, hardware integration, and mass production. Hands-on with ARM-based Qualcomm QCM6125 / Quectel SOM platforms, Android (AOSP), OTA, and high-speed multilayer PCB design in Altium. Proven at taking products from prototype to production, working shoulder-to-shoulder with cross-functional hardware teams, mentoring junior engineers, and building structured technical documentation.

TECHNICAL SKILLS

Languages: C, C++, Python, Embedded C, Bash/Shell, ARM Assembly, Verilog

Firmware & OS: Embedded Linux, AOSP/Android, BSP & device-driver integration, kernel customization, OTA (full & incremental), RTOS, Perfetto

Hardware & PCB: Altium Designer, EasyEDA, high-speed flex & multilayer PCB design, EMI control, signal integrity, schematic/layout review

Platforms & Protocols: Qualcomm QCM6125, Quectel SC668S SOM, ARM; I2C, SPI, UART

Production & QC: SMT oversight (reflow, X-ray, AOI, BGA dye test), GMS key flashing/provisioning, FCC compliance (EMI/EMC, ESD, OTA), validation & burn-in

Debug & Tools: Git, GDB, oscilloscope, logic analyzer, signal probing, QNavigator/QFlash/QPST, Proteus, MATLAB

EXPERIENCE

AIO APP Inc
Islamabad, Pakistan

Senior Hardware Design Engineer (Firmware)
11/2023 – Present

- **Firmware & BSP:** Spearheaded firmware development on the ARM-based Qualcomm QCM6125 platform; performed Linux kernel/BSP and AOSP customization, device-driver integration, and board bring-up across display, touch, camera, audio, and charging subsystems.
- **AOSP & OTA:** Customized AOSP and implemented full and incremental OTA update delivery on Qualcomm QCM6125 devices; also worked on the Android layer of off-the-shelf Rockchip-based devices.
- **Audio:** Performed microphone data processing, dual-mic noise cancellation, and speaker tuning for clear audio in compact designs.
- **Power & Charging:** Integrated battery calibration and charging ICs supporting fast and reverse wireless charging; implemented thermal cutoffs and safety logic.
- **PCB Design:** Designed high-speed flex and multilayer PCBs with EMI control and signal integrity; conducted detailed schematic and component-level reviews in Altium Designer.
- **Collaboration & Mentoring:** Resolved board-level issues with hardware teams using signal probing, oscilloscopes, and logic analyzers; mentored junior engineers and maintained thorough technical documentation.

AIO APP Inc
Shenzhen, China

International Engineering Visits — China (4 Visits)

- Across four engineering visits to AIO’s China manufacturing partner, progressed from device bring-up and optimization to full-scale mass production.
- **Bring-Up & Optimization:** Performed device bring-up and optimized key peripherals (display, touch, battery, speaker, MIC) for power efficiency and performance.
- **Validation & Compliance:** Led antenna tuning (2.4/5 GHz), FCC compliance (EMI/EMC, ESD, OTA), USB, vibration/drop testing, and thermal profiling with 8-probe monitoring.
- **SMT & Mass Production:** Oversaw SMT (footprints, reflow, X-ray, AOI, BGA dye testing) and drove the production line with GMS key flashing and provisioning, final firmware deployment, and functional testing through to product shipment.

TeReSol Pvt. Ltd
Islamabad, Pakistan

Hardware Design Engineer
08/2023 – 11/2023

- Embedded design and development in C/C++; developed Bash/Shell scripts for process automation.
- Performed bug tracing and fixing to deliver functional firmware; cloned and flashed customized NVIDIA Tegra (ARM/Linux) GPU cards.

PROJECTS

Tableside AI Device 11/2023 – Present

- Developed system firmware on the Quectel SC668S SOM, integrating display, touch, camera, and audio for real-time interaction; implemented battery management, fast charging, and reverse wireless charging; designed high-speed flex and multilayer PCBs with signal integrity and EMI control.

Semi-Autonomous Object-Tracking Sentry Robot (FYP) 06/2022 – 09/2022

- Built a detection, tracking, and aiming system on Raspberry Pi with Python; improved tracking efficiency and designed a double-layer PCB.

Image Convolution on FPGA 06/2022 – 09/2022

- Implemented optimized convolution algorithms in Verilog on a Xilinx FPGA with VGA output of source and result images.

EDUCATION

COMSATS University Islamabad Islamabad, Pakistan
BS Electrical (Computer) Engineering 09/2019 – 09/2023

CERTIFICATIONS

Advanced Embedded Linux Development Specialization — Coursera
Mastering Linux: The Complete Guide to Becoming a Linux Pro — Udemy
Internet of Things (IoT), Robotics and Hacking with NodeMCU — Udemy

AWARDS

- **Microwiz Winner —** Visospark ’23, COMSATS
- **Leadership Excellence Award —** IEEE SAC Leadership Dialogue 2022
- **Sumo Robot Winner —** HI ROBO TEC, Taxila

INTERESTS

Sports: Table Tennis, Badminton
Adventure: Traveling, Photography